

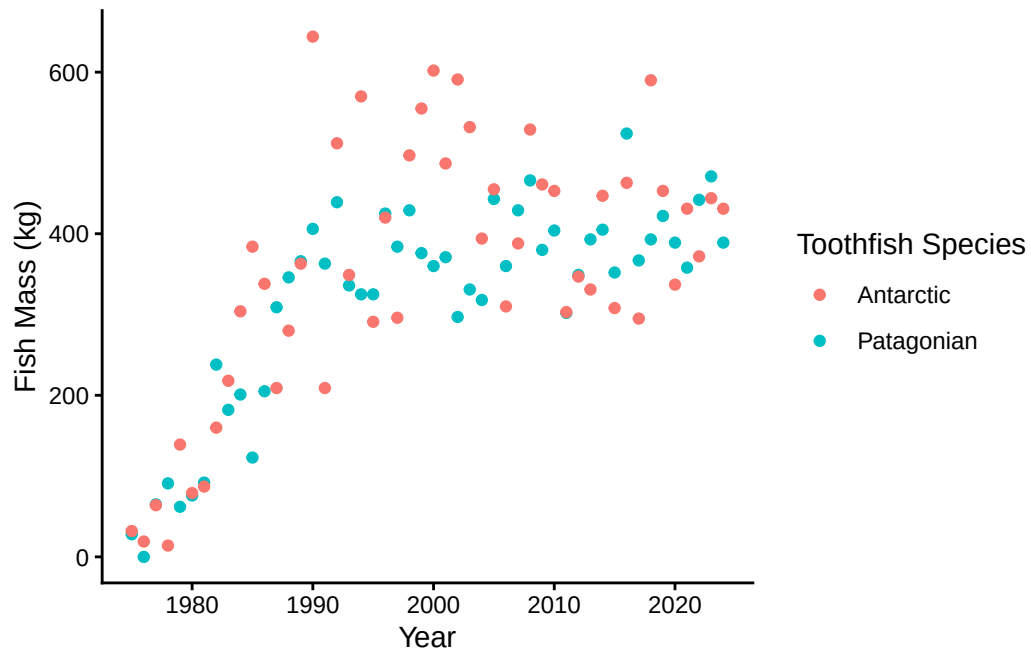
Module 5, Assignment 3: Output Check

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2.

```
# A tibble: 100 x 3
  species    fish_kg  date
  <chr>      <dbl> <dbl>
1 Patagonian    28  1975
2 Patagonian     0  1976
3 Patagonian    65  1977
4 Patagonian    91  1978
5 Patagonian    62  1979
6 Patagonian    76  1980
7 Patagonian    92  1981
8 Patagonian   238  1982
9 Patagonian   182  1983
10 Patagonian   201  1984
# i 90 more rows
```

4.



6.

```
# A tibble: 50 x 3
  species  fish_kg  date
  <chr>      <dbl> <dbl>
1 Patagonian    28  1975
2 Patagonian     0  1976
3 Patagonian    65  1977
4 Patagonian    91  1978
5 Patagonian    62  1979
6 Patagonian    76  1980
7 Patagonian    92  1981
8 Patagonian   238  1982
9 Patagonian   182  1983
10 Patagonian   201  1984
# i 40 more rows

# A tibble: 50 x 3
  species  fish_kg  date
  <chr>      <dbl> <dbl>
1 Antarctic    32  1975
2 Antarctic    19  1976
3 Antarctic    64  1977
```

```

4 Antarctic      14 1978
5 Antarctic     139 1979
6 Antarctic      79 1980
7 Antarctic      87 1981
8 Antarctic     160 1982
9 Antarctic     218 1983
10 Antarctic     304 1984
# i 40 more rows

```

7.

A 'drc' model.

Call:

```
drm(formula = fish_kg ~ date, data = pat, fct = LL.4())
```

Coefficients:

```

b:(Intercept)  c:(Intercept)  d:(Intercept)  e:(Intercept)
      -628.9632        -0.6173        391.1327        1983.6795

```

8.

A 'drc' model.

Call:

```
drm(formula = fish_kg ~ date, data = ant, fct = LL.4())
```

Coefficients:

```

b:(Intercept)  c:(Intercept)  d:(Intercept)  e:(Intercept)
      -719.317        -1.875        430.739        1983.039

```

10.

A tibble: 50 x 4

	species	fish_kg	date	predicted_values
	<chr>	<dbl>	<dbl>	<dbl>
1	Patagonian	28	1975	22.7
2	Patagonian	0	1976	30.8
3	Patagonian	65	1977	41.3
4	Patagonian	91	1978	54.8

5	Patagonian	62	1979	71.7
6	Patagonian	76	1980	92.3
7	Patagonian	92	1981	117.
8	Patagonian	238	1982	144.
9	Patagonian	182	1983	174.
10	Patagonian	201	1984	205.

i 40 more rows

A tibble: 50 x 4

	species	fish_kg	date	predicted_values
	<chr>	<dbl>	<dbl>	<dbl>
1	Antarctic	32	1975	20.2
2	Antarctic	19	1976	29.2
3	Antarctic	64	1977	41.5
4	Antarctic	14	1978	57.9
5	Antarctic	139	1979	79.2
6	Antarctic	79	1980	106.
7	Antarctic	87	1981	138.
8	Antarctic	160	1982	174.
9	Antarctic	218	1983	213.
10	Antarctic	304	1984	252.

i 40 more rows

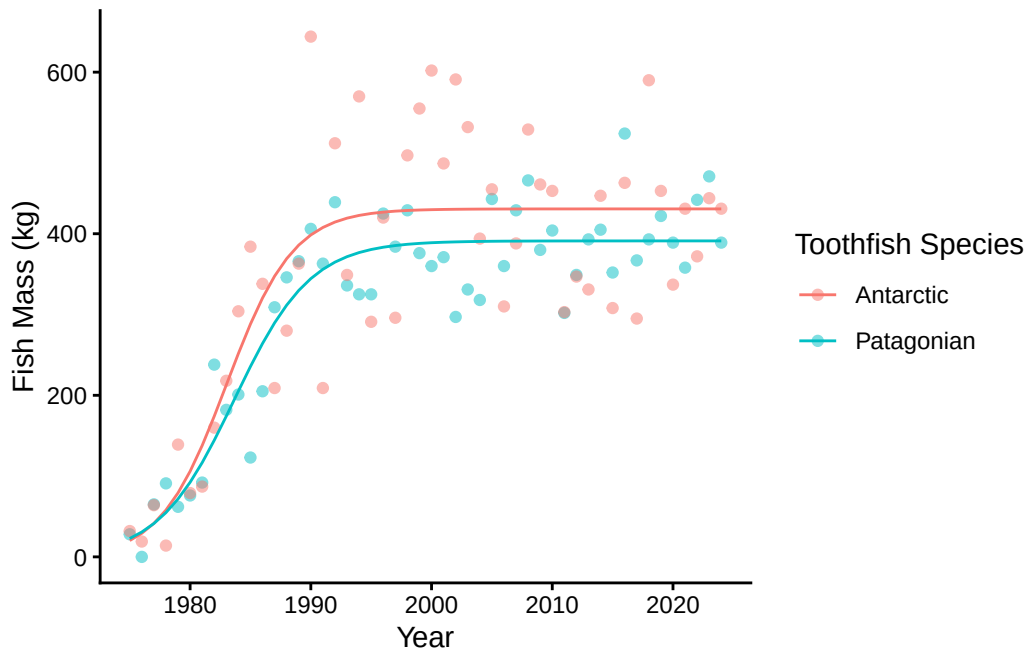
11.

A tibble: 100 x 4

	species	fish_kg	date	predicted_values
	<chr>	<dbl>	<dbl>	<dbl>
1	Patagonian	28	1975	22.7
2	Patagonian	0	1976	30.8
3	Patagonian	65	1977	41.3
4	Patagonian	91	1978	54.8
5	Patagonian	62	1979	71.7
6	Patagonian	76	1980	92.3
7	Patagonian	92	1981	117.
8	Patagonian	238	1982	144.
9	Patagonian	182	1983	174.
10	Patagonian	201	1984	205.

i 90 more rows

12.



13.

```
# A tibble: 60 x 4
  species    fish_kg  date predicted_values
  <chr>      <dbl> <dbl>          <dbl>
1 Patagonian    325  1995          381.
2 Patagonian    425  1996          383.
3 Patagonian    384  1997          385.
4 Patagonian    429  1998          387.
5 Patagonian    376  1999          388.
6 Patagonian    360  2000          389.
7 Patagonian    371  2001          389.
8 Patagonian    297  2002          390.
9 Patagonian    331  2003          390.
10 Patagonian   318  2004          390.
# i 50 more rows
```

14.

```
# A tibble: 2 x 3
  species    avg_fish_kg stdev_fish_kg
  <chr>      <dbl>          <dbl>
1 Antarctic    427.          94.9
```

2 Patagonian 388. 51.8

16.

Welch Two Sample t-test

data: fish_kg by species

t = 1.9562, df = 44.877, p-value = 0.05669

alternative hypothesis: true difference in means between group Antarctic and group Patagonian

95 percent confidence interval:

-1.147373 78.414040

sample estimates:

mean in group Antarctic mean in group Patagonian

427.1000

388.4667